Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 8

Disjunctive Normal Form

Covers Lehman, Leighton & Meyer, *MCS* Section 3.4.1

Status of this set

Ungraded practice. Solutions released with it. Attempt everything first. Fill truth tables on this page.

Part I: How to Practice This Material

Count first, write second

The recipe, and the check that goes with it

One AND-term per **true** row. In each term, a variable appears *plain* if it is T in that row and *negated* if it is F.

Before writing anything, count the T's in the column. That is exactly how many AND-terms you will have. If you finish with a different number, you have made an error — and you will know immediately rather than after checking every row.

Three mistakes to expect

1. **Building from the false rows.** That produces the CNF instead. False rows are Friday.
2. **Omitting a variable from a term.** In DNF every AND-term names every variable.
3. **Expecting DNF to be short.** It usually is not. Normal form means standard shape, not small.

Part II: Practice Problems

Section A — Reading DNF off a table

1. Build the truth table for $(\sim P) \vee Q$, then write its DNF.

P	Q	$(\sim P) \vee Q$	AND-term
T	T		
T	F		
F	T		
F	F		

2. Write the DNF of each. You may work from a table or reason directly.

(a) $P \wedge Q$

(b) $\sim Q$ (over the variables P and Q)

(c) $P \vee Q$

Section B — Three variables

3. Build the DNF of $(P \wedge Q) \vee R$.

P	Q	R	$(P \wedge Q) \vee R$	AND-term
T	T	T		
T	T	F		
T	F	T		
T	F	F		
F	T	T		
F	T	F		
F	F	T		
F	F	F		

Section C — Using DNF to test equivalence

4. Write the DNF of each formula, then say whether the two are equivalent.

$$\text{(i)}\ \ \sim(P \Rightarrow Q) \qquad\qquad \text{(ii)}\ P \wedge (\sim Q)$$

5. Two students propose conditions for granting access:

$$\text{Ana: } (P \wedge Q) \vee (P \wedge (\sim Q)) \qquad \text{Ben: } P \wedge (Q \vee (\sim Q))$$

Are they equivalent? Answer by comparing DNFs, and say in one sentence what both conditions really amount to.

Section D — Writing

6. In five to eight complete sentences, explain why every compound statement has a DNF, and why two formulas are equivalent exactly when their DNFs match. Refer to the recipe.

Section E — Challenge (optional)

7. For n variables, what is the largest possible number of AND-terms in a DNF? Which formulas achieve it, and what formula has the *fewest*?